CS341 · COMPUTER GRAPHICS

What Repeats 472 · 462 · 461

Questions repeated across the past midterms

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What's inside

01 The midterm format

02 Repeated in more than one model

03 Asked once, worth memorizing

04 Problems you must master

01 The midterm format

Part	472	462	461
Definitions and short answers	✓	✓	✓
True / False	✓	✓	✓
MCQ	✓	✓	✓
Fill in the blanks		✓	✓
A line-drawing problem (DDA or Bresenham)	DDA	Bresenham	Bresenham
CRT figure	✓		

02 Repeated in more than one model

Question	Where	Answer
Bresenham from (3,3) to (10,7)	462 · 461	(3,3) (4,4) (5,4) (6,5) (7,5) (8,6) (9,6) (10,7)
Absolute vs relative coordinates	472 · 461	Absolute = the actual position from the origin. Relative = an offset from the last position (the current position).
Aspect ratio (why / compute)	472 · 461	Keeps shapes intact: a square stays a square, a circle does not become an ellipse. $AR = \frac{\frac{Yp}{Ymax}}{\frac{Xp}{Xmax}}$
Define computer graphics	472 · 461	Creation, storage and manipulation of models and images
What does a CRT emit?	472 · 461	Electrons
"Applications are only for entertainment"	462 · 461	**False**
CG applications: CAD / medical / entertainment / ...	472 · 461	All of the above
Bresenham's decision variable	462 ×2	Picks the next pixel: (x+1, y) or (x+1, y+1)
Refresh rate	461 ×2	How many times per second the picture is redrawn

Raster vs random for drawing lines	472 · 461	Random scan draws lines directly with the beam

03 Asked once, worth memorizing

Word-for-word definitions

Term	Definition
Resolution	The maximum number of points displayed on a CRT without overlap
Persistence	The time for the emitted light to decay to one-tenth of its original intensity
Color gamut	All colors produced by mixing the primary colors
Tint	White pigment added to pure pigment
Tone	Both white and black added to pure pigment
Vector	Has magnitude and direction
Vector from P to Q	`v = Q − P`

True / false

Statement	Answer
A vector has both magnitude and direction.	**True**
Sum two points results in a valid point.	**False**
The dot product of two perpendicular vectors is equal to zero.	**True**
A unit vector has a length greater than 1.	**False**
M + Y + C = White.	**False**
Raster scan scans the screen from vertical to horizontal.	**False**
Bresenham's line algorithm uses floating-point arithmetic.	**False**
In raster displays, pixels are arranged in a rectangular grid.	**True**

Colors

`Yellow + Magenta = Red` `Magenta + Cyan = Blue` `Green + Blue = Cyan`

`Digital displays → RGB` `Subtractive mixing → CMY(K)`

Fixed MCQ answers

Question	Answer
Where is the raster picture stored?	Frame buffer
Screen clarity depends on?	Dot pitch
Which is an input device?	Mouse
NOT a main task?	Data encryption
Goal of CG	Synthetic images indistinguishable from reality

04 Problems you must master

1. Bresenham's line (asked twice with the same numbers). Memorize: `p0 = 2Δy − Δx`; negative → `+2Δy`; zero or positive → `y+1` and `+2Δy − 2Δx`.
2. DDA, especially with slope above 1 (y increases by 1 and x by $\frac{1}{m}$).
3. Computing the aspect ratio.
4. The CRT figure with letters.

Model 462 has questions on B-Reps, voxels, CSG, BSP and rotation about a point, which are not in the current chapter slides.